

Long Questions:**TOTAL: 80 marks****Question 11****[5]**

A town is flooded with water and a rescue pilot is moving at 15 m/s parallel to the flat ground 100 m above the ground to airdrop food. How far, in the x direction, should the plane be from the target if a sack of flour released from the plane is to strike the target?

Question 12**[5]**

- Determine the range, in the x direction, of the shell that is being fired from a gun with a muzzle velocity v at an angle θ .
- Determine the angle of elevation θ of a gun that fires a shell with a muzzle velocity of 120 m/s and hits a target on the same level, but 1300 m downrange from the initial position.

Question 13**[5]**

A block of mass $m_1 = 3.7$ kg on frictionless plane inclined at an angle of $\theta = 30^\circ$ is connected by a cord over a massless, frictionless pulley to a second block of mass $m_2 = 2.3$ kg, as shown in Figure 1. What are

- the magnitude of the acceleration of each block?
- the direction of the acceleration of the hanging block?
- the tension in the cord?

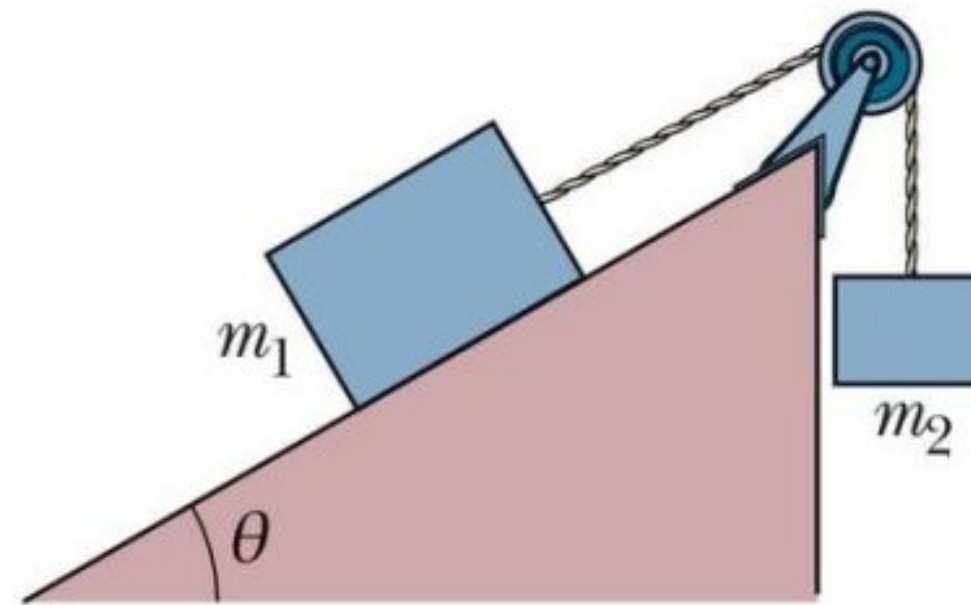


Figure 1: Question 13

Question 14**[5]**

During a storm, a crate of lobster is sliding across a slick, oily parking lot through a displacement $\mathbf{d} = -3 \text{ m } \mathbf{i}$ while a steady wind pushes against the crate with a force of $\mathbf{F} = (2\text{N}) \mathbf{i} + (-6\text{N}) \mathbf{j}$. The situation and coordinate axes are shown in Figure 2.

- How much work does this force do on the crate during the displacement?
- If the crate has a kinetic energy of 10 J at the beginning of the displacement $\mathbf{d}$, what is the kinetic energy at the end of $\mathbf{d}$?

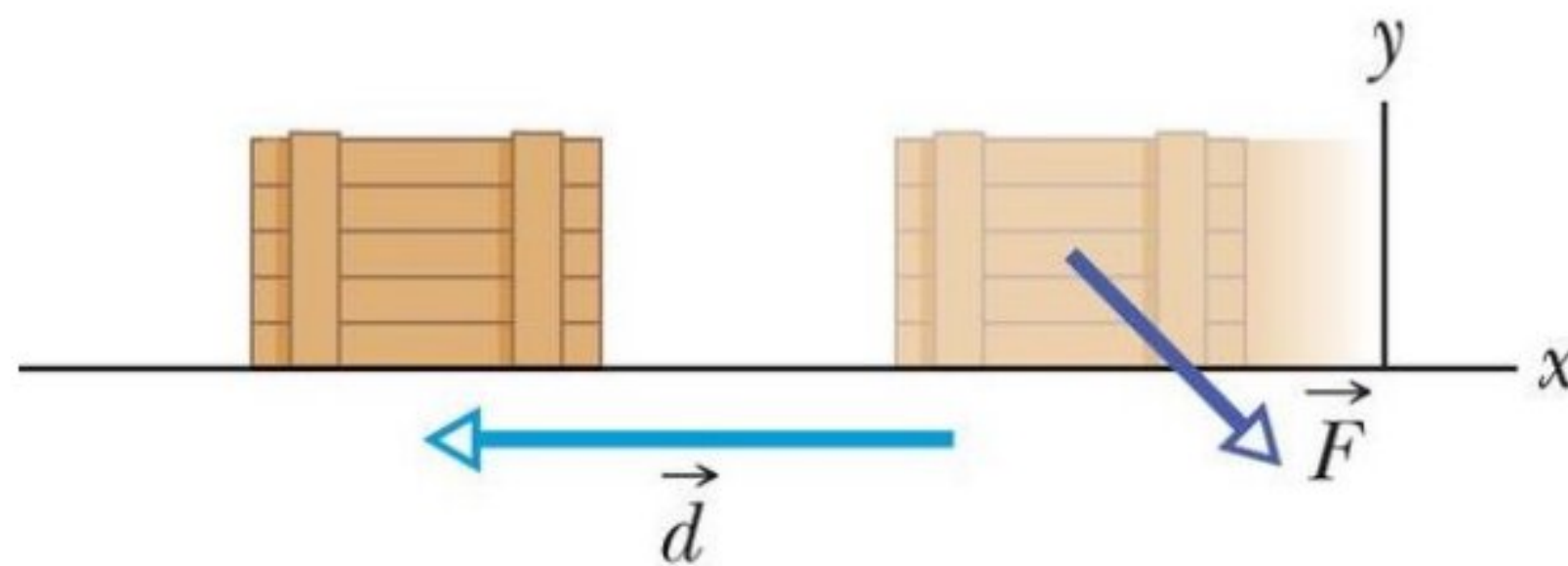


Figure 2: Question 14

Question 15**[5]**

Figure 3 shows two industrial spies sliding an initially stationary 225 kg floor safe a displacement $\mathbf{d}$ of magnitude 8.5 m. The push $\mathbf{F}_1$ of spy 001 is 12 N at an angle of 30° downward from the horizontal; the pull $\mathbf{F}_2$ of spy 002 is 10 N at 40° above the horizontal. The magnitudes and directions of these forces do not change as the safe moves, and the floor and safe make frictionless contact. What is the net work done on the safe by the forces $\mathbf{F}_1$ and $\mathbf{F}_2$ during the displacement $\mathbf{d}$?

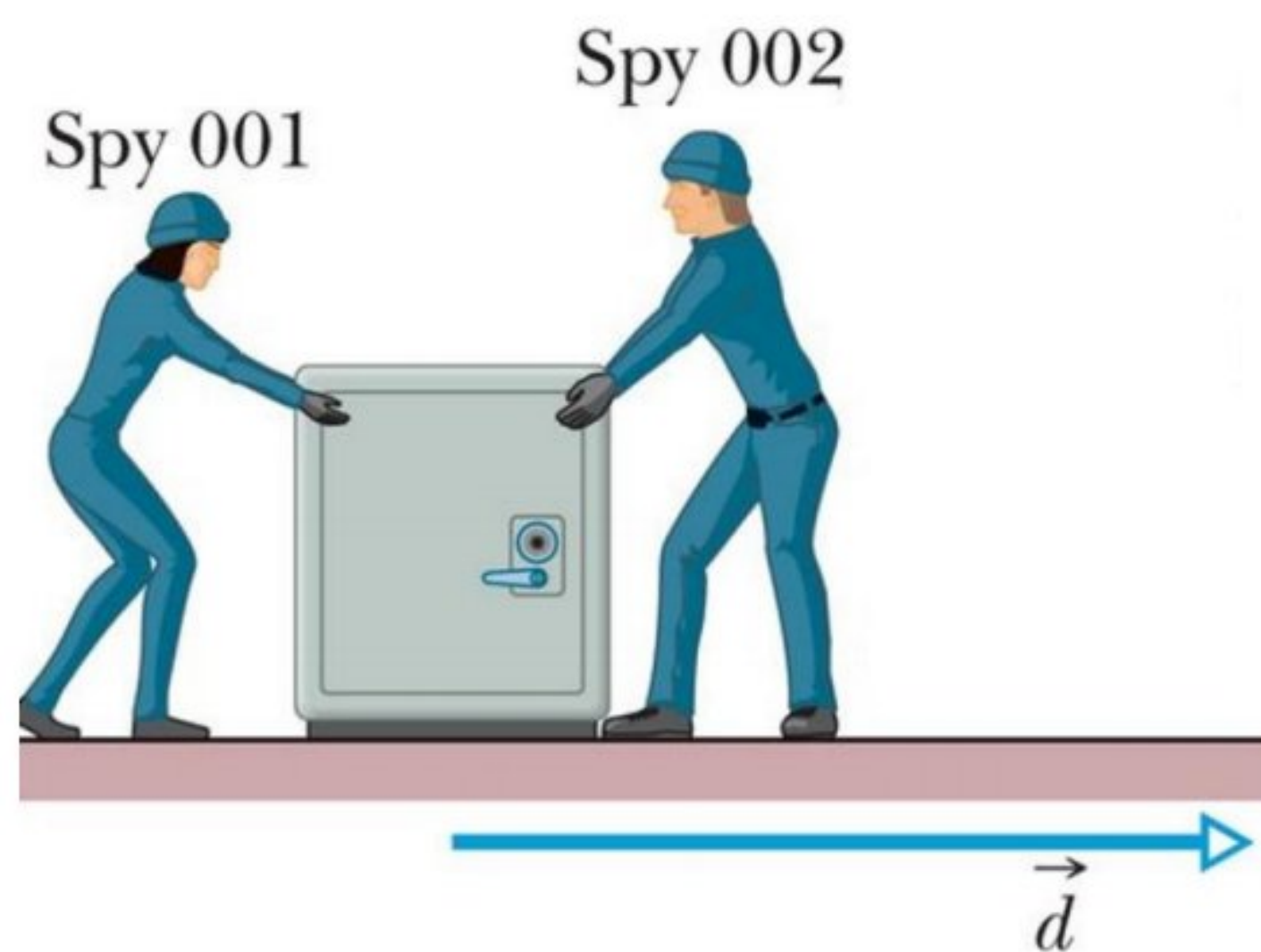


Figure 3: Question 15

Question 16**[5]**

In Figure 4, force $\mathbf{F}$ accelerates block m_1 to the right. Find its acceleration in terms of F and the coefficient of friction, μ_k , at the contact surface.

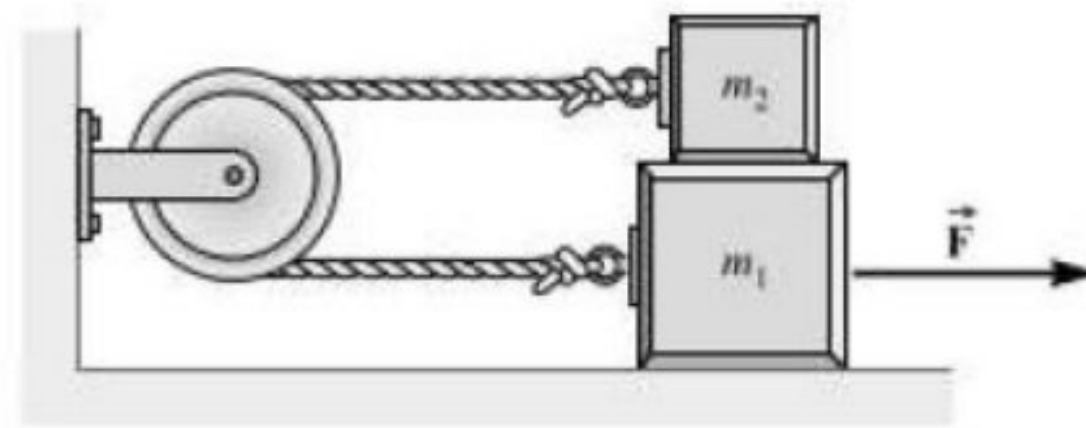


Figure 4: Question 16

Question 17**[5]**

As shown in Figure 5, a 15 g bullet is fired horizontally into a 3 kg block of wood suspended by a long cord. The bullet sticks in the block. Compute the speed of the bullet if the impact causes the block to swing 10 cm above the initial level.

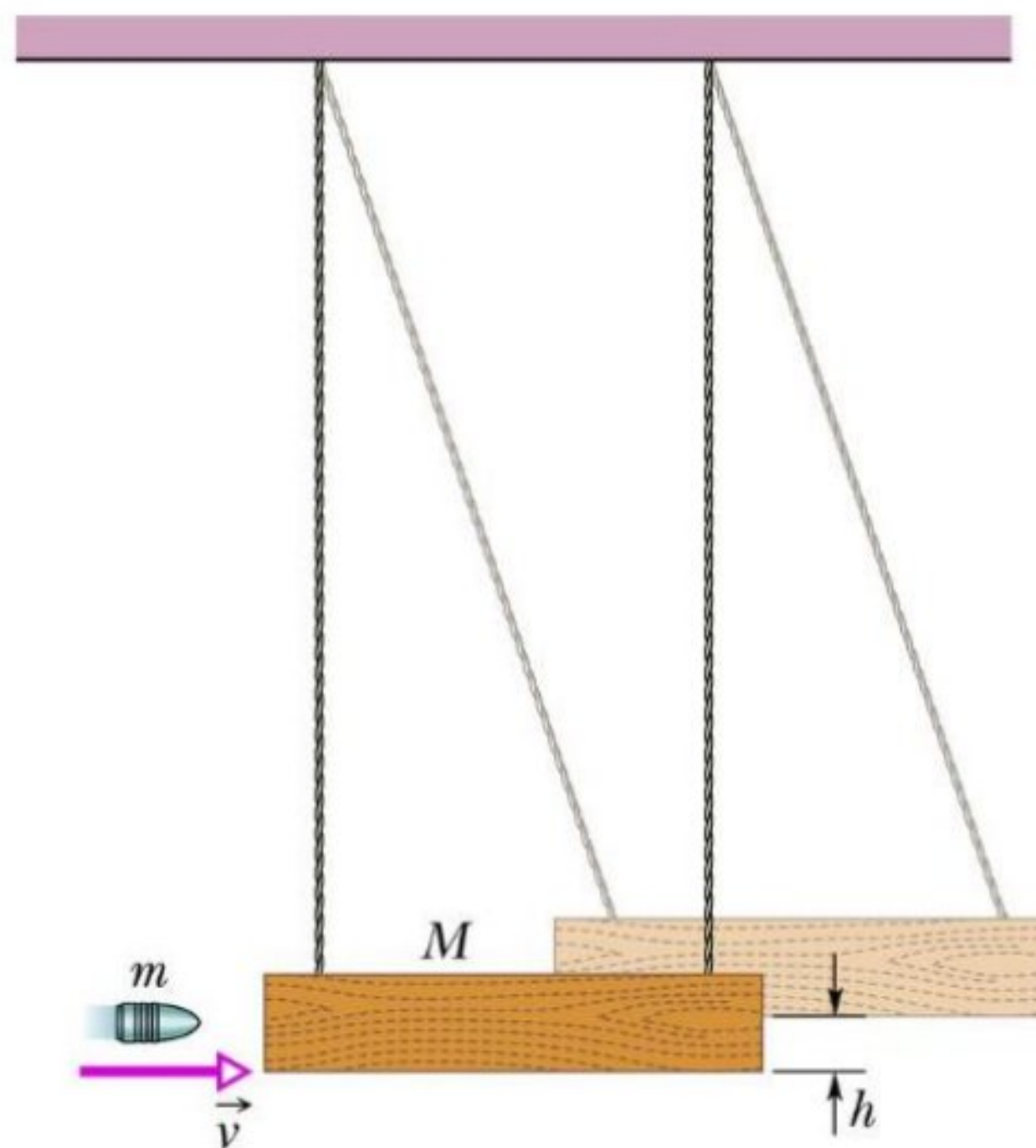


Figure 5: Question 17

Question 18**[5]**

A child of mass M is released from rest at the top of a water slide of height 8.5 m above the bottom of the slide. Assuming the slide is frictionless because of the water on it, find the speed of the child at the bottom of the slide.

Question 19**[5]**

The water tank in Figure 6 has a hole in it close to the bottom of the tank. Only using symbols, what is the speed v of the water exiting the water tank? If the height difference between the top of the water level inside the tank and the hole is 10 m , calculate the speed of the water exiting the water tank.

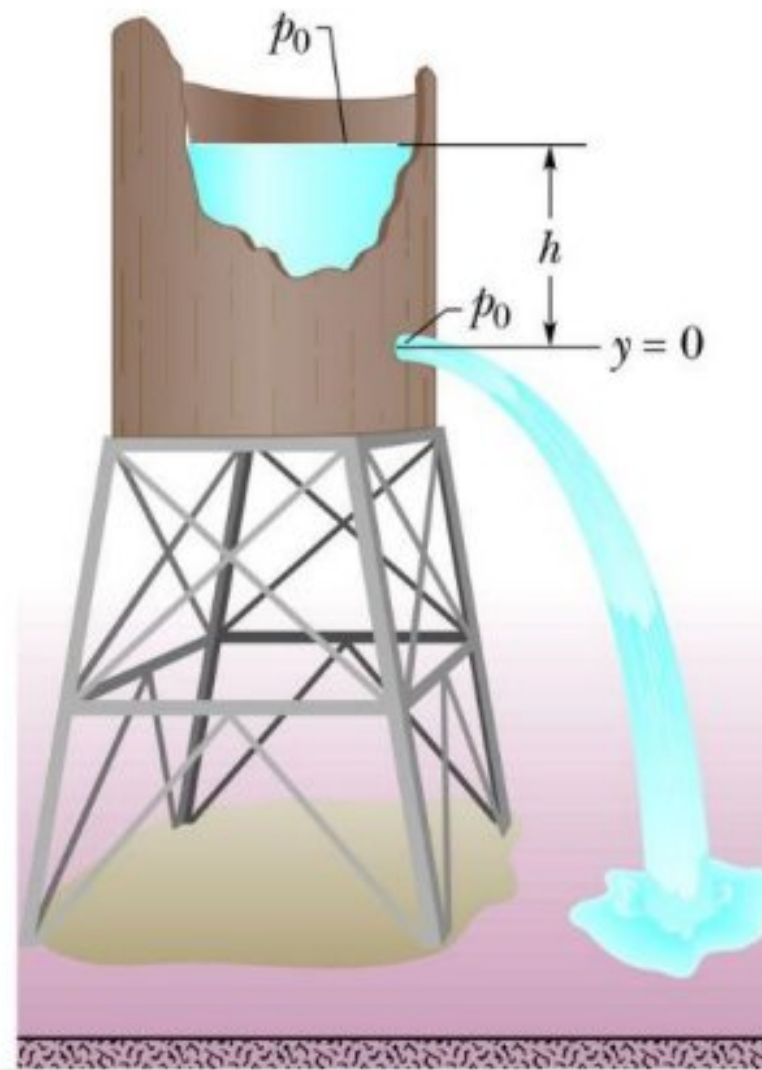


Figure 6: Question 19

Question 20**[6]**

Proof that the kinetic energy of a rigid body in pure rotation is given by $K = \frac{1}{2}I\omega^2$.

Question 21**[6]**

Derive an equation for the pressure in a fluid at a specific point as a function of depth.

Question 22**[7]**

The pipe shown in Figure 7 has an area 0.020 m^2 at point 1 and $7.85 \times 10^{-3}\text{ m}^2$ at point 2. At point 1 the pressure is 200 kPa , and oil of density 800 kg m^{-3} flows through the pipe at a rate of $0.03\text{ m}^3\text{ s}^{-1}$. Calculate the pressure at point 2 if viscous effects are ignored.

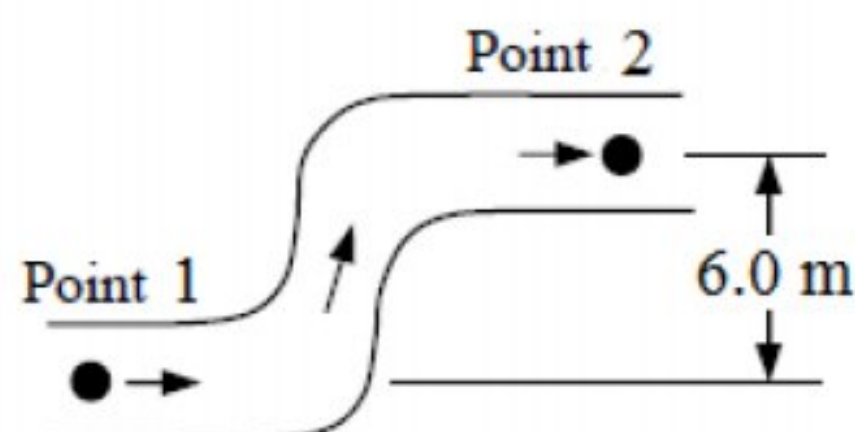


Figure 7: Question 22

Question 23**[9]**

A wave travelling along a string is described by: $y(x, t) = (0.0351\text{m})\sin[(52.3 \text{ rad/s})x + (2.52 \text{ rad/s})t]$ with x in meters and t in seconds.

- a) What is the wavelength of the wave?
- b) What is the period of oscillation?
- c) What is the frequency of the wave?

Question 24**[4]**

A Carnot engine operates between 235°C and 115°C , absorbing $6.30 \times 10^4 \text{ J}$ per cycle at the higher temperature.

- a) What is the efficiency of the engine?
- b) How much work per cycle is this engine capable of performing?

Question 25**[3]**

One mole of oxygen (assume it to be an ideal gas) expands at a constant temperature T of 310 K from an initial volume V_i of 12 L to a final volume of V_f of 19 L . How much work is done by the gas during the expansion?

TOTAL/TOTAAL: 100